

Dreaming of Butterflies: Embodying Zhuangzi's Philosophy through a Mixed Reality Installation

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Figure 1: Spatial visualization of the immersive environment, which blends physical reality with a virtual ink-painting aesthetic, creating a seamless integration of the real and the imagined.

Abstract

While extended reality (XR) has increasingly been used for narrative immersion and cultural visualization, few projects have explored its potential to embody eastern metaphysical thoughts through spatial interactions. In our work *Dreaming of Butterflies*, we address this

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gap by transforming Chinese philosopher Zhuangzi's classic parable, the Butterfly Dream, into an interactive XR journey that invites participants to experience the boundaries between illusion and reality, self and other. By merging embodied interaction, stylized visual language, and Daoist cosmology, the experience demonstrates how XR can serve as a medium for metaphysical reflection and poetic experience, offering a novel framework for spatializing philosophical thought in immersive art.

CCS Concepts

• **Human-centered computing** → *Mixed / augmented reality*.

Keywords

XR Interaction, Embodied Cognition, Philosophical Narrative, Daoist Design, Metaphorical Space

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1 Introduction

Immersive extended reality (XR) technologies have enabled increasingly complex forms of spatial storytelling. However, despite technical advancements, XR applications often prioritize visual immersion and interaction over conceptual depth. Many narrative-driven XR experiences focus on simulation, gamified engagement, or aesthetic spectacle, with relatively few exploring the medium's capacity for philosophical inquiry. While early immersive artworks such as Char Davies's *Osmose* demonstrated how metaphysical themes could be expressed through embodied navigation and visual abstraction, these cases remain exceptions [Davies and Harrison 1996]. The integration of philosophical discourse into XR environments, particularly in ways that are experiential and spatial rather than discursive, remains underexplored within both artistic and technological domains.

One philosophical framework well suited to spatialization is the classical Daoist parable *Butterfly Dream*, attributed to Chinese philosopher Zhuangzi. In this allegory, the philosopher dreams he is a butterfly and, on waking, wonders whether he is a man who dreamed of being a butterfly or a butterfly now dreaming he is a man. The story articulates key Daoist concerns, including the indeterminacy of illusion and reality, the relativity of identity, and the dissolution of binary oppositions, which invite a reconsideration of how reality is structured and how subjectivity emerges. The parable withholds resolution and cultivates contemplative uncertainty. These concepts provide fertile ground for immersive multisensory expression in XR, yet few projects to date attempt to render such metaphysical questions as embodied experience.

In our work *Dreaming of Butterflies*, we address this gap by translating the philosophical tension of Zhuangzi's *Butterfly Dream* into an XR installation that unfolds across mixed and virtual reality spaces. Guided by a butterfly, participants move through a spatial journey consisting of a physical MR entry scene and three symbolic VR environments, with the MR scene staged as a real tea-room installation whose tangible furnishings and ambience are integral to the experience. Each stage corresponds to a facet of Daoist thought: spontaneity, perspectival transformation, and the dissolution of self. The project draws on the visual language of traditional Chinese ink painting, using hand-crafted textures, stylized shaders, and ambient soundscapes to evoke a poetic sensibility. Interaction is non-instructive and intuitive: users are not told what to do, but discover through presence, opting for the Daoist concept of effortless action. Building on previous XR works that fuse cultural narrative and immersion, such as *LanternOperAR* [Xu et al. 2025] and *Sunghaksipdo VR* [Lee and Lee 2024], which incorporate Confucian and Neo-Confucian philosophy through interactive environments, *Dreaming of Butterflies* distinguishes itself by treating philosophy not as content to be illustrated, but as a perceptual structure to be inhabited. In doing so, the work positions XR as a platform not

just for storytelling or simulation, but for embodied metaphysical reflection.

2 Related Work

A growing body of research has explored how XR can be used to blur the boundaries between real and virtual environments. For instance, George et al. proposed a design framework for seamless transitions across the mixed reality continuum, allowing users to fluidly change perspectives without breaking immersion [George et al. 2020]. Similarly, another work *Nature: Metaphysics + Metaphor* by Braud et al. demonstrated how MR environments could evoke intangible concepts, such as impermanence and duality, through spatial composition and poetic interactivity [Braud et al. 2024]. These projects foreground fluidity and continuity as essential to maintaining narrative coherence across hybrid spaces.

Other works have extended XR to the domain of philosophical or abstract reflection. For instance, *Sunghaksipdo VR* transforms a Neo-Confucian diagram into a participatory learning experience, allowing users to explore cosmological relationships through embodied navigation [Lee and Lee 2024]. Similarly, *LanternOperAR* integrates Yangmingist philosophy into an augmented opera experience, using tangible props and floating virtual inscriptions to immerse users in cultural and ethical introspection [Xu et al. 2025]. These projects illustrate how immersive systems can serve not only as vessels for storytelling, but as interactive platforms for intellectual and spiritual exploration.

Within the realm of aesthetic engagement, several XR works have drawn on traditional East Asian visual traditions. *InkBrush* introduces a sketch-based painting interface that mimics the expressive irregularity of ink wash brushstrokes in real-time 3D space [Yao et al. 2024]. Meanwhile, *Illustrating Ten Poems* reconstructs classical Chinese landscapes as interactive environments, translating poetic imagery into spatial form [Cao et al. 2023]. These works demonstrate how XR can preserve, reinterpret, and revitalize traditional aesthetics in novel and interactive ways.

Based on these precedents, *Dreaming of Butterflies* integrates philosophical embodiment, MR–VR transitions, and stylized visuals into a unified experience. Rather than representing philosophy, it is embodied through spatial ambiguity, perceptual shifts, and open-ended interaction. This positions immersive systems not only as narrative stages but also as epistemological tools for engaging with metaphysical thought.

3 Interaction Design

Dreaming of Butterflies constructs a multi-stage XR narrative inspired by Zhuangzi's *Butterfly Dream*, where boundaries between dream and reality, self and other, remain unstable. Using MR and VR as metaphors for perceptual and ontological transformation, the design immerses participants in liminal environments where boundaries blur and identity is fluid.

The journey follows a cyclical structure, commencing and concluding in the same physical MR tea room. Users progress through three symbolic VR environments, referred to as “chapters”, each embodying core Daoist concepts. The spatial-narrative sequence and associated interactions are visualized in Figure 2. The butterfly serves as a constant guide across these chapters, linking each

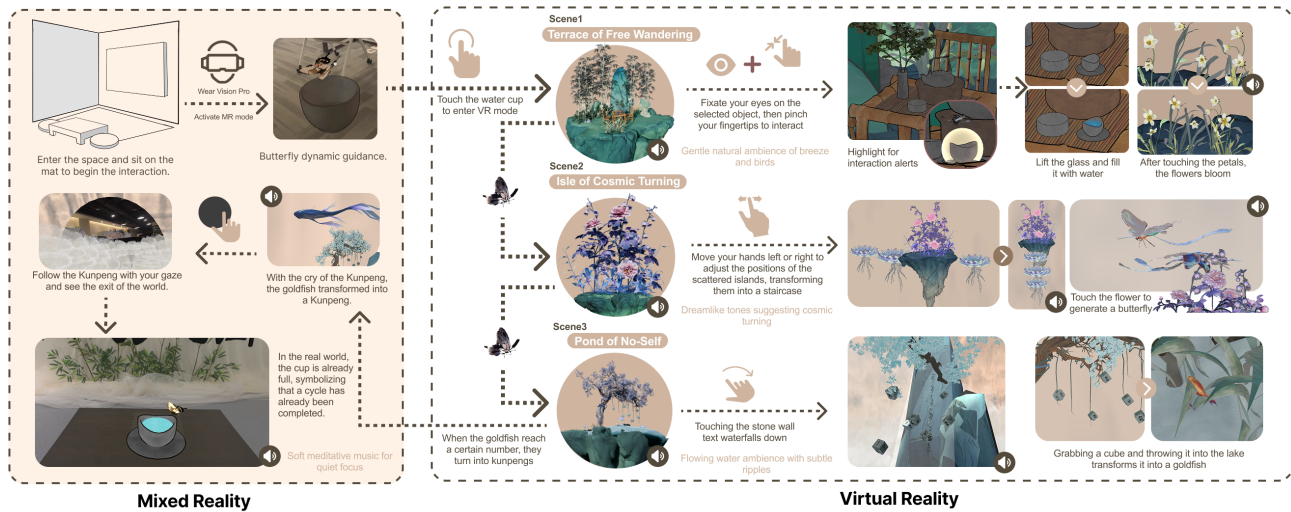


Figure 2: Overview of the spatial narrative and interaction flow, illustrating the participant's journey from the physical MR tea room through three symbolic VR environments—each facilitating a distinct Daoist theme via gesture-based interaction—before returning to reality, completing a cyclical experiential loop.

philosophical insight to a distinct interactional form. Inspired by XR works such as Embodied Movement and Generative Media Performance [Wendell and Guan 2024], which structure embodied interactivity as a narrative arc, we adopt a chapter-based approach to highlight the relational development between stages and reinforce the poetic rhythm of the participant's transformation.

3.1 Chapter 1: MR Tea Room

The experience begins in a serene mixed reality tea room anchored in the participant's physical space. In this space, a delicate butterfly flutters in and lands atop a simple teacup on the table. A soft golden halo of ink-like light ripples outward from the cup, a visual cue inviting interaction. The ambience is quiet save for gentle background audio, setting a contemplative mood of perceptual ambiguity. This initial scene deliberately makes the participant question what is real or illusory, foreshadowing the dream-like journey to come, which echoes Zhuangzi's reflections on reality and illusion. When the participant reaches out and touches the cup, the mundane tea room transforms, ushering the participant from MR into the first of three symbolic VR islands.

3.2 Chapter 2: Terrace of Free Wandering

The first VR chapter transports the participant to a tranquil floating island that continues the tea room's dreamlike atmosphere. A virtual bamboo grove surrounds a familiar tea table, sustaining spatial continuity and softening the boundary between physical and digital. Warm light, soft breezes, and distant windchimes evoke a contemplative mood. Interactions embody the Daoist notion of *wuwei* (effortless action): when the participant's gaze lingers on a closed flower bud, it gradually unfurls; a subtle pinch-and-release motion toward an empty teacup conjures a gentle stream of tea, as if poured by an unseen helper. As interaction deepens, calligraphic

script emerges across the sky bearing the line, "There are great beauties in heaven and earth that are not spoken of, and the great Daoist road has nothing to do with them." The butterfly reappears, circling gently, inviting the participant toward the next transformation.

3.3 Chapter 3: Isle of Cosmic Turning

The butterfly leads into a more surreal environment where three large lotus platforms hover in midair. A central radiant bloom is visible but unreachable. The ambient soundscape deepens to heighten mystery. Interactivity centers on rotation as participants spin the platforms with hand gestures and align them into a path toward the bloom. The spatial puzzle enacts perspectival relativism, where coherence emerges from fragmentation only when the participant shifts their point of view. This interaction draws on Zhuangzi's *Qi Wu Lun* ("Equalizing Things"), emphasizing that truth depends on relational framing rather than absolutes. Upon touching the bloom, a massive butterfly emerges and ascends, leaving the line "There is nothing that is not the other, and nothing that is not the self." The participant then follows this guide to the final island.

3.4 Chapter 4: Cliff of No-Self

The third island sets no fixed goals. The participant enters a twilight landscape with a circular pond, a central tree bearing luminous text blocks, and a high cliff. Interactions are metaphorical rather than goal-oriented: touching the wall reveals a waterfall of flowing script; casting a text block into the pond generates a golden fish. As these fish accumulate and orbit, the atmosphere shifts: light pulses, sound deepens, and surrounding forms shimmer. The fish then converge into the silhouette of a Kunpeng, a Daoist creature symbolizing boundless transformation, which ascends in a burst of mist and ink. Unlike previous chapters, this space withholds clarity and invites an experiential meditation on the impossibility of fixed

meaning, where meaning arises from transformation rather than analysis.

3.5 Chapter 5: Return

As the Kunpeng disappears into the sky, a rift opens above the virtual landscape and reveals the MR tea room. The butterfly reappears near the breach, circling. The participant reaches toward the soft glow to return. Familiar sounds come back with water boiling and a faint wind. The tea room seems unchanged, yet subtle differences appear as the space feels slightly warmer and quieter. The butterfly perches on the same cup, now gently steaming with tea. Overhead, a final inscription reads, “Am I Zhuangzi who dreamt of being a butterfly, or a butterfly now dreaming I am Zhuangzi?”, then it fades. The work closes on recurrence rather than resolution, leaving participants in ambiguity after a journey where spatial design, interactivity, and Daoist cosmology converge.

4 Technical Implementation

4.1 Interaction System

The interaction design of *Dreaming of Butterflies* centers on embodied, gesture-driven engagement implemented through Unity’s XR Interaction Toolkit and PolySpatial framework, optimized for Apple Vision Pro’s spatial capabilities. Natural hand tracking removes controllers and supports the work’s emphasis on spontaneity and presence. A unified gesture vocabulary spans mixed and virtual reality, so that reaching, pinching, and gaze elicit symbolic responses such as flowers blooming, cups filling, and platforms rotating without explicit instruction.

Narrative progression is guided by a custom-designed Butterfly Guidance Mechanism, which acts as both spatial navigator and symbolic conductor. This virtual butterfly dynamically responds to user interaction by pausing near key objects, visually marked by particle halos, and advancing only when the user completes the associated task. This structure encodes implicit pacing into the experience, mirroring the Daoist principle of “spontaneous yet guided unfolding.” The butterfly’s smooth Bezier-spline path and contextual appearance reinforce the dreamlike continuity of the experience.

To ensure coherent state transitions, a centralized logic controller monitors user actions, manages environmental changes, and synchronizes the appearance of floating inscriptions derived from Zhuangzi’s writings. These inscriptions emerge only upon interaction completion, reinforcing a contemplative rhythm and linking action to philosophical revelation. Together, these elements constitute a cohesive interaction system in which user agency, narrative structure, and philosophical meaning are tightly interwoven through embodied participation.

4.2 Stylized Rendering Pipeline

The visual design of *Dreaming of Butterflies* draws inspiration from the aesthetics of traditional Chinese ink painting, employing a custom real-time stylization pipeline implemented using Unity Shader Graph, which is the only shader authoring method supported by Vision Pro. All 3D assets were modeled in Blender and textured with hand-painted ink-style maps created in Photoshop (Figure 3).

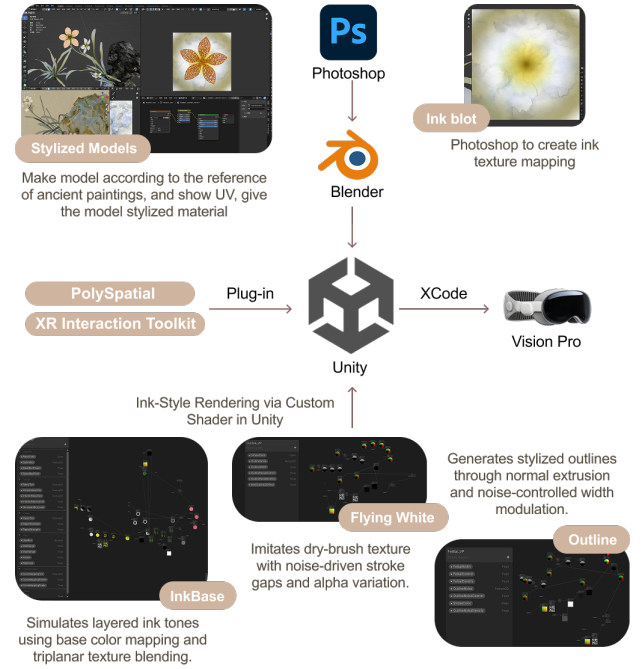


Figure 3: Technical pipeline illustrating the integration of stylized assets, ink-inspired shaders, and gesture-based interaction within the Unity environment for Apple Vision Pro.

These textures were then processed through a dual-material shader system, combining a base color pass with stylized edge rendering.

To replicate the expressive irregularity of ink strokes, noise-modulated alpha masks are used to simulate *feibai* (hollow strokes), a technique characterized by dry-brush texture and tonal discontinuity often found in unfinished brushwork. *Moyunran* (ink diffusion) is recreated using layered Gaussian blur and triplanar blending, mimicking the bleed of pigment on absorbent rice paper. Surface detail is further enhanced by paper texture overlays modulated by surface normals, suggesting a fibrous physical substrate.

Edge contours are stylized through procedural outline detection with noise-based thickness variation to produce hand-drawn silhouettes. Color is processed with a custom ink-monochrome LUT and ACESFilm tone mapping, yielding a subdued analog composition. This stylization reinforces the experience’s philosophical undertone by foregrounding ambiguity, fluidity, and the porous boundary between form and void, thereby echoing Zhuangzi’s worldview through visual abstraction.

5 Reflection

5.1 Daoist Embodiment and Philosophical Spatialization

Our project explores how XR can embody Daoist philosophy as lived, spatial practice rather than abstract content. Zhuangzi’s Daoism values *wuwei* (effortless action), *shifei* (perspectival transformation), and *ziran* (natural spontaneity), encouraging openness and responsiveness [Li 2025]. In our project, *wuwei* is illustrated

through an unprompted touch that makes a flower bloom; *shifei* is enacted by aligning drifting platforms, creating coherence through shifting perspectives; and *ziran* is conveyed through textual blocks transforming into a Kunpeng, showing identity as process rather than essence. Within this embodiment framework, our work serves as a space for practicing Daoist sensibilities, where the butterfly subtly guides, goals remain suggestive, and feedback is partial, allowing meaning to emerge through the user's interaction with the space.

Previous XR works have drawn on philosophy or culture by presenting symbols or texts for users to decode, such as Sunghaksipdo VR recreating a Neo-Confucian diagram in 3D [Lee and Lee 2024], or LanternOperAR overlaying Yangmingist inscriptions on AR props [Xu et al. 2025]. In contrast, Dreaming of Butterflies uses XR to perform metaphysics through direct experience, without overt diagrams or written philosophy. Participants engage with ambiguity, aligning more with Osmose [Davies and Harrison 1996], which used a dreamlike landscape to induce unity with nature. Like Osmose, our design involves the whole body to shift perception, but while Osmose relies on abstract metaphors, Dreaming of Butterflies embeds Daoist themes into its mechanics and narrative. Philosophical meaning emerges through exploration of liminal spaces, echoing Zhuangzi's view that wisdom arises in the flow of life.

5.2 Audience Feedback

Our work was exhibited at the Art Exhibition Center of Communication University of China. An informal user study involving 20 participants—many without prior VR/AR experience—demonstrated its accessibility: the majority completed the experience with minimal instruction, facilitated by the intuitive, metaphorical butterfly-guided interaction flow. Audience responses revealed some interesting preferences. For instance, some participants appreciated the three-island, chaptered design for its clarity, while others desired greater freedom to explore. We view these perspectives as complementary: the linear structure offers legibility and pacing, whereas the open design fosters emergent experiences and self-directed meaning. Additionally, some participants also commented that the brief philosophical statements presented after each interaction were not always easy to understand, suggesting that the poetic ambiguity designed to invite reflection can occasionally lead to confusion. Future iterations could explore ways to gently bridge this gap, such as providing optional contextual guidance for those who seek it, without compromising the open-ended nature of the experience.

5.3 Limitations and Future Work

Despite demonstrating the potential of XR to convey Daoist philosophy, there are some limitations in the current design. First, the chapter-based structure provides a coherent and well-paced journey but limits free exploration. Future versions can investigate nonlinear structures and branching metaphors to balance narrative guidance with open discovery. Furthermore, the brief philosophical sentences that appear after each interaction can be difficult to interpret. We treat these texts function as reflective prompts rather than final statements. Building on this strategy, we will incorporate adaptive interaction patterns that enable the system to respond dynamically to user hesitation, pacing, and curiosity. Finally, the

present sound design, while incorporating ambient music and local audio cues for each scene, may not fully sustain immersion. To address this, we will expand spatialized audio and introduce adaptive multimodal layers such as olfactory and haptic cues, and evaluate their impact through user testing. As XR platforms evolve, new capabilities in gaze estimation, affective sensing, and language-model-supported narrative generation will further enrich this form of poetic interactivity.

6 Conclusion

Dreaming of Butterflies demonstrates how XR can serve as a vessel for abstract philosophical inquiry, not merely through symbolism but through lived, spatialized experience. The work reframes mixed and virtual reality as spaces not only for visual spectacle but for internal transformation. By making Daoist thought accessible through sensory and gestural engagement, the piece invites viewers to question binaries of self and other, dream and reality. It contributes to a growing body of practice-based research at the intersection of immersive technology and cultural philosophy, affirming that meaningful technological art must be both experientially rich and conceptually rigorous.

Acknowledgments

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